Choosing the Promoter-Mining Ensemble

How six DNA models were compared, and why three were deployed

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Abstract

We evaluated six machine-learning models for finding bacterial promoters in the genome of our target, *Clostridium butyricum*, and chose the best three for the deployed ensemble. The headline result is that a model's score on a standard *balanced* test is a poor guide to how it behaves at *genome scale* — the setting that actually matters when mining a real genome. On two held-out *Clostridium* genomes, a DNA foundation-model probe (Evo2), a fine-tuned language model (ProkBERT), and a custom fusion CNN were the three most precise, and combining them into one ensemble beat every individual model. Two Nucleotide Transformer variants and a simpler CNN were evaluated and set aside.

1 What we measured

Six models were each tested two ways. The first is a **balanced test**: equal numbers of real promoters and non-promoters, the standard machine-learning benchmark (scored by the area under the precision–recall curve). The second is a **genome-scale test**, which slides each model across an entire genome held out of training, where real promoters make up only a tiny fraction of all positions — the realistic mining setting.

For the genome-scale test we report **precision at a fixed recall**. *Recall* is the fraction of real promoters the model finds; *precision* is the fraction of its predictions that are correct. So "precision at 50% recall" reads: when the model is tuned to recover half of the real promoters, what fraction of the sites it flags are genuine? Higher is better — it means fewer false alarms. We use two *Clostridium* genomes held out of training, *C. ljungdahlii* and *C. autoethanogenum*, as close-relative stand-ins for the target.

The six models are a DNA foundation-model probe (**Evo2**: a lightweight classifier on top of the Evo2 model from the Arc Institute, which is used as-is, not retrained), a fine-tuned prokaryotic language model (**ProkBERT**), a custom convolutional network trained from scratch that fuses several views of the sequence (**Fusion CNN**), a plainer from-scratch convolutional network as a control (**Simple CNN**), a DNA transformer fine-tuned with LoRA (**Nucleotide Transformer**), and a larger DNA transformer that reads a wider window (**Nucleotide Transformer v3**).

2 Balanced accuracy is not genome-scale performance

On the balanced test the ranking is almost the *reverse* of what matters at genome scale. Only one model clears 0.90 — the Nucleotide Transformer v3, at 0.96 (Table 1)[1] — and it is the *weakest* of the six at genome scale (Table 2). The Evo2 probe is its mirror image: *lowest* of all on the balanced test (0.70), yet the *strongest* at genome scale. This reversal is the central lesson of the evaluation, and the reason we chose the deployment models on genome-scale behaviour rather than balanced accuracy.

The genome-scale ranking is stable across both genomes: Evo2 leads, ProkBERT and the Fusion CNN follow closely, and the two Nucleotide Transformers and the simple CNN trail — the exact opposite of the balanced-test order, where Evo2 is last.

[1] Not directly comparable to the others: v3 reads a wider input window than the 81-base-pair models, which inflates its balanced score.

Table 1: Balanced-test score (area under the precision–recall curve; higher is better). Only the Nucleotide Transformer v3 clears 0.90 — and Table 2 shows the genome-scale ranking is nearly the reverse.

Model	Balanced-test score
Nucleotide Transformer v3	0.960
ProkBERT	0.884
Nucleotide Transformer	0.854
Fusion CNN	0.848
Simple CNN	0.804
Evo2 probe	0.703

Table 2: Genome-scale precision on two held-out *Clostridium* genomes, at 10% and 50% recall. Higher is better (fewer false positives). ✓ marks the three models chosen for the deployed ensemble.

		C. ljungdahlii		*C. autoethanogenum*	
Model	Deployed	@10%	@50%	@10%	@50%
Evo2 probe	✓	**71.0%**	**58.0%**	**85.8%**	**71.4%**
ProkBERT	✓	64.4%	47.7%	76.7%	64.6%
Fusion CNN	✓	64.4%	47.2%	81.0%	66.0%
Simple CNN	—	49.0%	40.3%	72.6%	63.0%
Nucleotide Transformer	—	43.9%	36.1%	64.9%	58.7%
Nucleotide Transformer v3	—	35.7%	28.0%	58.5%	48.3%

3 The ensemble beats every single model

Rather than pick one model, we combine the top three by agreement: each ranks every candidate site, and we average those ranks. Because it depends only on the ordering of sites, this combination is unaffected by the models scoring on different scales. The result is the most precise caller on both held-out genomes.

Table 3: The three-model ensemble versus its members: genome-scale precision at 10% and 50% recall.

	C. ljungdahlii		*C. autoethanogenum*	
	@10%	@50%	@10%	@50%
Three-model ensemble	**74.4%**	**61.6%**	**87.0%**	**76.6%**
Evo2 probe	69.4%	58.4%	87.6%	72.5%
ProkBERT	64.4%	47.7%	76.7%	64.6%
Fusion CNN	64.4%	47.2%	81.0%	66.0%

The ensemble matches or beats the best single model at every recall level, and is decisively ahead at 50% recall on both genomes. Adding a fourth model made it *worse* on both (for example, precision at 50% recall on *C. ljungdahlii* fell from 61.6% to 58.3%), confirming that three complementary models is the right number.[2]

4 What we deployed

The final ensemble is three deliberately different models — one DNA foundation model (Evo2), one language model (ProkBERT), and one convolutional network (the Fusion CNN) — so their independent strengths reinforce one another. This ensemble was then used to mine the target genome, *C. butyricum*, producing a ranked shortlist of candidate promoters for experimental validation.

[2]Single-model figures differ by about a percentage point between Tables 2 and 3 because they come from two separate evaluation runs; the comparisons *within* each table are the valid ones.